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School of Computer Science



M.Sc. (Information Technology) - Batch 2025- 27

SOCS010201

Object Oriented Programming

Project Title: Anand Sewa NGO

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Abstract

This project is made for a NGO name "Anand Sewa" to help them manage donation records in a digital way. Before this project, the NGO people were writing all donation details in a big paper register book. But sometime paper gets lost, or rain water falls on it and makes it wet, so the data gets destroyed very easily. In this project, we use Java programming language and Swing framework to make nice buttons, text boxes, and windows. People can login with email and password, and then fill their name and money amount. There is also a special Admin Dashboard where the boss man can see all records in a list, and he can add new one, update the old one, or delete them if there is any mistake. We keep all the data in an ArrayList which is like a big memory box in the computer. We learn a lot of things in this project like how to make classes, objects, and event handling. It is very good for the real world and helps NGO people work very easy and fast without any headache.

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Introduction

1.a. Objective of the Project

The main objective of this project is to make a good and working software for Anand Sewa NGO. It help to manage all the donation records in a digital way so everything is clean. The goal is to stop using paper work because paper work is very slow and can be lost very easily. When we use paper, sometime the pages get torn or coffee spill on it and the writing become unreadable. By using this software, the NGO can save alot of time and keep data safe in the computer memory forever. Also, it make the NGO look very professional and modern in front of the donors. When donors see a computer system, they trust more that their money is going to right place. This project also help us students to learn how to make real world application that solve actual problems of people. We want to make a system that is easy to use even for old people who don't know computer very well, so we make big buttons and clear text. The main thing is to make donation process smooth and happy for everyone involved.

1.b. Overview of Java Programming

sJava is a very good and strong programming language that is used all over the world. It was made by James Gosling and it have a coffee cup logo which is very nice. Java have objects and classes which is called Object Oriented Programming, or OOP for short. It is very safe and can run on many different types of computers like Windows, Mac, or Linux because it have a thing called JVM which make it run anywhere. In this project, we use Java with the Swing library to make a Graphical User Interface (GUI). GUI mean we can make windows with buttons and text fields instead of just black screen with white text.

In Java, a class is like a blueprint of a house, and an object is the actual house we build from that blueprint. For example, we make a Donation class, and then we make many Donation objects for each person who give money. Java also have a thing called Garbage Collection, which is like a garbage man that come and clean up the memory when we don't need things anymore, so the computer don't get slow. We also use Exception Handling in Java, which mean if something go wrong, like user type letters instead of numbers, the program don't just crash and die, but it show a nice error message instead. This make Java very robust and good for making big projects like this NGO system. We choose Java over other languages because it is very strict and teach us good programming habits, and the Swing library is already there so we don't need to download extra things to make buttons and windows.

2. Project Requirements

2.a. Problem Statement

The NGO people was doing manual entry of donations in a physical register book. Sometime they forget the donor names or the exact money amount because human brain make mistakes.

The data get lost or damaged by water, fire, or insects. Also, when they want to find a old record from last year, they have to flip hundred of pages which take long time and make them tired.

This project solve this specific problem by saving all the information securely in a computer ArrayList so it never get lost and we can find it in one second.

2.a.i. Describe the specific problem addressed in the project

- Manual data entry is very slow and make human mistakes like writing wrong numbers.
- Physical books take too much space in the office and need a big cupboard.
- Finding old records in a big book is very hard and take long time.
- If the register book is lost or stolen, all the NGO data is gone forever and no one can recover it.

2.b. Technical Requirements

We need some software and hardware to run this project good and make sure it don't hang.

2.b.i. Software and hardware prerequisites

- **Software:** Java Development Kit (JDK) must be installed on the computer to compile the code.
- **Software:** A code editor like Notepad, Eclipse, or IntelliJ IDEA to write the Java code.

- **Hardware:** Any normal working computer or laptop is fine.
- **Hardware:** It should have at least 4GB RAM (which is the short term memory of computer) and 256GB storage to run the Java program smoothly without freezing.

Design and Architecture

3.a. Algorithm/Workflow Explanation

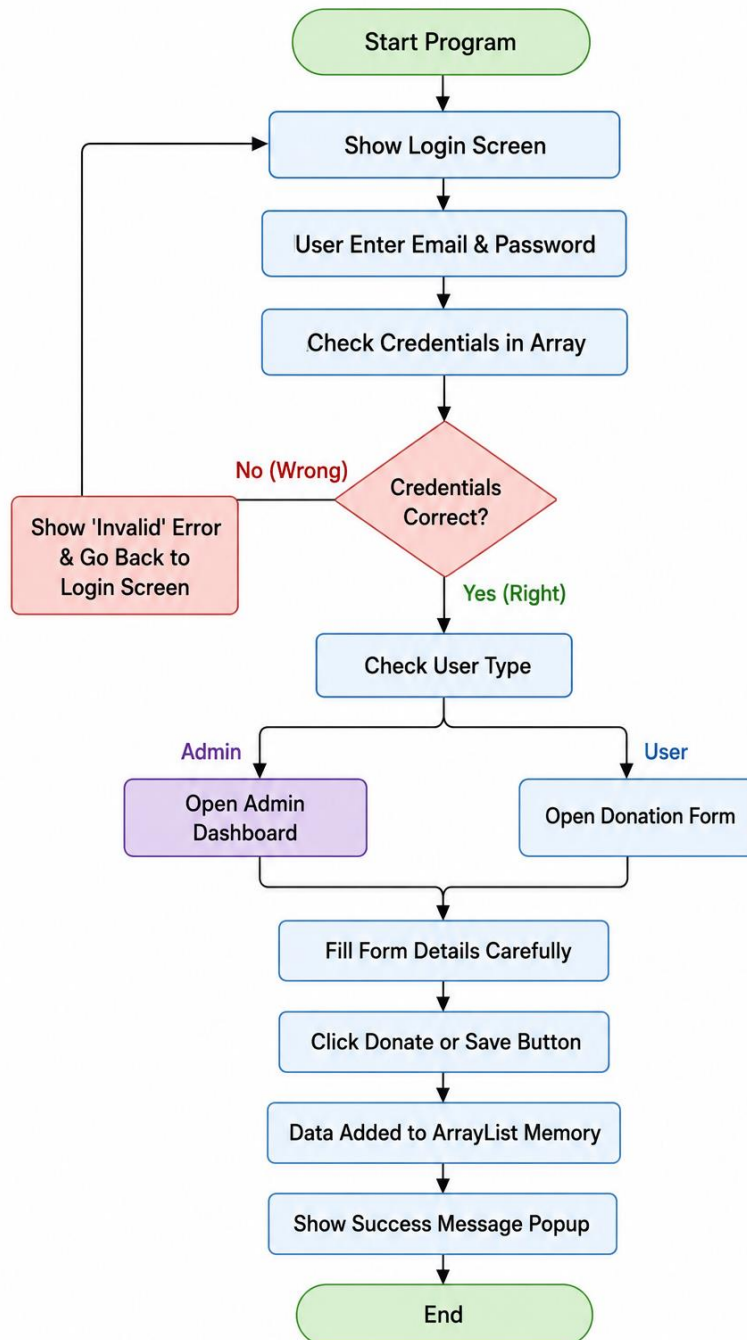
First, the user open the program and the Login Screen come on the monitor. The user type their email and password in the empty boxes using keyboard. The program check if the email and password is right by comparing it with the list inside the code. If it is wrong, it show a red error message popup and ask to try again. If it is right, it check if the user is Admin or Normal user. If Admin, it go to Admin Dashboard screen where he can see everything. If Normal, it go to Donation Form screen. The user fill all details like name, amount, and category, then click the Donate button with mouse. The data is save in the ArrayList and show a green success message saying thank you.

3.a.i. Detail the step-by-step approach used for solving the problem

- **Step 1:** Make the Login window with JFrame and add text fields for email and password.
- **Step 2:** Make the Donation class to hold data like a blueprint with all the variables.
- **Step 3:** Make the DonationManager class with a static ArrayList to store all the data objects together.
- **Step 4:** Make the HomeScreen for normal users to fill the donation form with many fields.
- **Step 5:** Make the AdminDashboard to see the list, add new, update old, and delete wrong records.

- **Step 6:** Connect all buttons with ActionListener so they do work when the user click them.

3.b. Flowchart



Implementation

4.1 Code Description

We write the code in small parts called classes so it is easy to understand and manage for everyone.

4.a.i Modular explanation of program

The program is divide into different classes and each class do one specific job only. The MyFrame class only do the login screen work. The HomeScreen class only do the donation form work. The AdminDashboard class only handle the list of donations. This make the code very clean and not messy, and if one part break, the other parts still work good. This is the main benefit of Object Oriented Programming.

4.a.ii. Discuss functions, control flow, arrays, structures, and commands used

- We use ArrayList which is a dynamic array to store many Donation objects because normal array size cannot change but ArrayList can grow big.
- We use if-else control flow to check if the password is correct or if the text fields are empty.
- We use for loop to check all the users in the array during the login process to find a match.
- We use functions like refreshList() to update the screen visually when the data change in the background.
- We use try-catch blocks for exception handling so if user type something weird, the program catch the error and don't crash.

4.b. Key Features and Commands

We use many cool Java commands to make the GUI work good and look nice.

4.b.i. Highlight major Java features utilized

- **JTextField:** We use this for typing text like name and email in a single line.
- **JComboBox:** We use this for selecting options like State or Category from a drop-down list so user don't type wrong spelling.
- **JButton:** We use this for clicking actions like "Donate Now" or "Logout".
- **JOptionPane:** We use this to show popup messages for errors or success, which is very easy to make.
- **ActionListener:** This is a very important feature that listen for mouse clicks and tell the program to do something when button is pressed.

4.c. Challenges in Implementation

Making the program was not always easy for me and I face many problems.

4.c.i. Discuss any specific implementation hurdles

- Making the GUI look nice and aligned was very hard because the components go everywhere.
- The GridLayout and BorderLayout managers was confusing sometime and I don't understand it at first.
- Sometime it make the text boxes too small or too big and the text get cut off.

- Connecting the button click to the ArrayList was tricky at first because I forget how to get the text from the text field.
- The image path for the logo was giving "file not found" error at first because the folder name was wrong.

4.d. Code:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.util.ArrayList;

class Donation {
    String id, donorName, email, phone, gender, city, state, amount, category,
        paymentMethod, pan, address, message, date;

    Donation(String id, String donorName, String email, String phone, String gender,
        String city, String state, String amount, String category, String paymentMethod,
        String pan, String address, String message) {
        this.id = id;
        this.donorName = donorName;
        this.email = email;
        this.phone = phone;
        this.gender = gender;
        this.city = city;
        this.state = state;
        this.amount = amount;
        this.category = category;
        this.paymentMethod = paymentMethod;
        this.pan = pan;
        this.address = address;
        this.message = message;
        this.date = java.time.LocalDate.now().toString();
    }
}

class DonationManager {
    static ArrayList<Donation> donations = new ArrayList<>();
}

// ADMIN DASHBOARD
class AdminDashboard extends JFrame {
    private DefaultListModel<String> listModel = new DefaultListModel<>();
```

```

private JList<String> donationList;

AdminDashboard() {
    setIconImage(new
ImageIcon("C:\\xampp\\htdocs\\practicefolder\\ANANDSEWA\\LOGO.png").getImage());
    setTitle("Admin Dashboard - Anand Sewa NGO");
    setSize(1100, 700);
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setLayout(new BorderLayout());

    JLabel title = new JLabel("DONATION RECORDS", SwingConstants.CENTER);
    title.setFont(new Font("Arial", Font.BOLD, 24));
    title.setBackground(new Color(30, 23, 96));
    title.setForeground(Color.WHITE);
    title.setOpaque(true);
    add(title, BorderLayout.NORTH);

    donationList = new JList<>(listModel);
    add(new JScrollPane(donationList), BorderLayout.CENTER);

    JPanel btnPanel = new JPanel();
    JButton addBtn = new JButton("Add Donation");
    JButton updateBtn = new JButton("Update");
    JButton deleteBtn = new JButton("Delete");
    JButton logoutBtn = new JButton("Logout");

    btnPanel.add(addBtn);
    btnPanel.add(updateBtn);
    btnPanel.add(deleteBtn);
    btnPanel.add(logoutBtn);
    add(btnPanel, BorderLayout.SOUTH);

    addBtn.addActionListener(e -> addDonationForm());
    updateBtn.addActionListener(e -> updateDonationForm());
    deleteBtn.addActionListener(e -> deleteDonation());
    logoutBtn.addActionListener(e -> { dispose(); new MyFrame(); });

    refreshList();
    setVisible(true);
}

private void refreshList() {
    listModel.clear();
    for (Donation d : DonationManager.donations) {
        listModel.addElement(d.id + " | " + d.donorName + " | ₹" + d.amount + " | " + d.category);
    }
}

```

```

private void addDonationForm() {
    showDonationForm(null, false);
}

private void updateDonationForm() {
    int index = donationList.getSelectedIndex();
    if (index == -1) {
        JOptionPane.showMessageDialog(this, "Please select a record to update");
        return;
    }
    showDonationForm(DonationManager.donations.get(index), true);
}

private void showDonationForm(Donation existing, boolean isUpdate) {
    JPanel panel = new JPanel(new GridLayout(0, 2, 10, 8));
    panel.setBorder(BorderFactory.createEmptyBorder(10, 10, 10, 10));

    JTextField nameField = new JTextField(existing != null ? existing.donorName : "");
    JTextField emailField = new JTextField(existing != null ? existing.email : "");
    JTextField phoneField = new JTextField(existing != null ? existing.phone : "");
    JTextField cityField = new JTextField(existing != null ? existing.city : "");
    JTextField amountField = new JTextField(existing != null ? existing.amount : "");
    JTextField panField = new JTextField(existing != null ? existing.pan : "");
    JTextArea addressArea = new JTextArea(existing != null ? existing.address : "", 3, 25);
    JTextArea messageArea = new JTextArea(existing != null ? existing.message : "", 3, 25);

    JComboBox<String> genderBox = new JComboBox<>(new String[]{"Male", "Female", "Other"});
    JComboBox<String> stateBox = new JComboBox<>(new String[]{"Punjab", "Gujarat", "Delhi", "Haryana", "Maharashtra", "Rajasthan"});
    JComboBox<String> categoryBox = new JComboBox<>(new String[]{"Education Support", "Healthcare", "Food Distribution", "Disability Support", "Women Empowerment", "Child Welfare"});
    JComboBox<String> paymentBox = new JComboBox<>(new String[]{"UPI", "Credit Card", "Debit Card", "Net Banking", "Cash"});

    if (existing != null) {
        genderBox.setSelectedItem(existing.gender);
        stateBox.setSelectedItem(existing.state);
        categoryBox.setSelectedItem(existing.category);
        paymentBox.setSelectedItem(existing.paymentMethod);
    }

    panel.add(new JLabel("Donor Name")); panel.add(nameField);
    panel.add(new JLabel("Email")); panel.add(emailField);
    panel.add(new JLabel("Phone")); panel.add(phoneField);
    panel.add(new JLabel("Gender")); panel.add(genderBox);

```

```

panel.add(new JLabel("City")); panel.add(cityField);
panel.add(new JLabel("State")); panel.add(stateBox);
panel.add(new JLabel("Amount")); panel.add(amountField);
panel.add(new JLabel("Category")); panel.add(categoryBox);
panel.add(new JLabel("Payment Method")); panel.add(paymentBox);
panel.add(new JLabel("PAN")); panel.add(panField);
panel.add(new JLabel("Address")); panel.add(new JScrollPane(addressArea));
panel.add(new JLabel("Message")); panel.add(new JScrollPane(messageArea));

String title = isUpdate ? "Update Donation" : "Add New Donation";
int result = JOptionPane.showConfirmDialog(this, panel, title,
JOptionPane.OK_CANCEL_OPTION);

if (result == JOptionPane.OK_OPTION) {
    String id = isUpdate ? existing.id : "D" + (DonationManager.donations.size() + 1001);
    Donation d = new Donation(id, nameField.getText(), emailField.getText(),
        phoneField.getText(), (String)genderBox.getSelectedItem(), cityField.getText(),
        (String)stateBox.getSelectedItem(), amountField.getText(),
        (String)categoryBox.getSelectedItem(), (String)paymentBox.getSelectedItem(),
        panField.getText(), addressArea.getText(), messageArea.getText());

    if (isUpdate) {
        DonationManager.donations.set(DonationManager.donations.indexOf(existing), d);
    } else {
        DonationManager.donations.add(d);
    }
    refreshList();
}

private void deleteDonation() {
    int index = donationList.getSelectedIndex();
    if (index != -1 && JOptionPane.showConfirmDialog(this, "Delete this record?" ==
JOptionPane.YES_OPTION) {
        DonationManager.donations.remove(index);
        refreshList();
    }
}

//HOME SCREEN
class HomeScreen extends JFrame {
    String u_email, userID, u_name;
    JTextField donorName, phone, city, amount, pan;
    JComboBox<String> stateBox, categoryBox, paymentBox;
    ButtonGroup genderGroup;
    JTextArea addressArea, messageArea;

```

```

JCheckBox anonymous;
String originalName;

HomeScreen(String u_name, String u_email, String userID) {
    this.u_email = u_email;
    this.userID = userID;
    this.u_name = u_name;
    this.originalName = u_name;

    setIconImage(new
ImageIcon("C:\\xampp\\htdocs\\practicefolder\\ANANDSEWA\\LOGO.png").getImage());

    Color primary = new Color(30, 23, 96);
    Color secondary = new Color(255, 172, 12);

    setTitle("Anand Sewa NGO - Donation Form");
    setExtendedState(JFrame.MAXIMIZED_BOTH);
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setLayout(new BorderLayout());

    // HEADING
    JLabel heading = new JLabel("ANAND SEWA NGO DONATION PORTAL");
    heading.setHorizontalAlignment(SwingConstants.CENTER);
    heading.setFont(new Font("Arial", Font.BOLD, 28));
    heading.setOpaque(true);
    heading.setBackground(primary);
    heading.setForeground(Color.WHITE);
    heading.setPreferredSize(new Dimension(1000, 70));

    // FORM PANEL
    JPanel formPanel = new JPanel(new GridLayout(0, 2, 15, 15));
    formPanel.setBorder(BorderFactory.createEmptyBorder(20, 30, 20, 30));
    formPanel.setBackground(Color.WHITE);

    formPanel.add(new JLabel("User ID"));
    JTextField txtID = new JTextField(userID); txtID.setEditable(false); formPanel.add(txtID);

    formPanel.add(new JLabel("Donor Name"));
    donorName = new JTextField(u_name); formPanel.add(donorName);

    formPanel.add(new JLabel("Email"));
    JTextField email = new JTextField(u_email); formPanel.add(email);

    formPanel.add(new JLabel("Phone Number"));
    phone = new JTextField(); formPanel.add(phone);

    formPanel.add(new JLabel("Gender"));

```



```

JPanel genderPanel = new JPanel(new FlowLayout(FlowLayout.LEFT));
genderPanel.setBackground(Color.WHITE);
JRadioButton male = new JRadioButton("Male");
JRadioButton female = new JRadioButton("Female");
JRadioButton other = new JRadioButton("Other");
genderGroup = new ButtonGroup();
genderGroup.add(male); genderGroup.add(female); genderGroup.add(other);
genderPanel.add(male); genderPanel.add(female); genderPanel.add(other);
formPanel.add(genderPanel);

formPanel.add(new JLabel("City"));
city = new JTextField(); formPanel.add(city);

formPanel.add(new JLabel("State"));
stateBox = new JComboBox<>(new String[]{"Punjab", "Gujarat", "Delhi", "Haryana", "Maharashtra", "Rajasthan"});
formPanel.add(stateBox);

formPanel.add(new JLabel("Donation Amount"));
amount = new JTextField(); formPanel.add(amount);

formPanel.add(new JLabel("Donation Category"));
categoryBox = new JComboBox<>(new String[]{"Education Support", "Healthcare", "Food Distribution", "Disability Support", "Women Empowerment", "Child Welfare"});
formPanel.add(categoryBox);

formPanel.add(new JLabel("Payment Method"));
paymentBox = new JComboBox<>(new String[]{"UPI", "Credit Card", "Debit Card", "Net Banking", "Cash"});
formPanel.add(paymentBox);

formPanel.add(new JLabel("PAN Number"));
pan = new JTextField(); formPanel.add(pan);

// Address
JPanel addressPanel = new JPanel(new BorderLayout());
addressPanel.setBackground(Color.WHITE);
addressPanel.setBorder(BorderFactory.createTitledBorder("Address"));
addressArea = new JTextArea(3, 20); addressArea.setLineWrap(true);
addressArea.setWrapStyleWord(true);
addressPanel.add(addressArea);

// Message
JPanel messagePanel = new JPanel(new BorderLayout());
messagePanel.setBackground(Color.WHITE);
messagePanel.setBorder(BorderFactory.createTitledBorder("Purpose / Message"));

```

```

        messageArea      =      new      JTextArea(3,      20);      messageArea.setLineWrap(true);
messageArea.setWrapStyleWord(true);
        messagePanel.add(messageArea);

// Options
JPanel optionPanel = new JPanel();
optionPanel.setBackground(Color.WHITE);
anonymous = new JCheckBox("Donate Anonymously");
JCheckBox newsletter = new JCheckBox("Receive NGO Updates");
anonymous.setBackground(Color.WHITE);
newsletter.setBackground(Color.WHITE);
optionPanel.add(anonymous);
optionPanel.add(newsletter);

anonymous.addActionListener(e -> {
    if (anonymous.isSelected()) {
        donorName.setText("Anonymous");
        donorName.setEditable(false);
    } else {
        donorName.setText(originalName);
        donorName.setEditable(true);
    }
});

// Buttons
JPanel buttonPanel = new JPanel();
buttonPanel.setBackground(Color.WHITE);
JButton donateBtn = new JButton("Donate Now");
JButton resetBtn = new JButton("Reset Form");
JButton logoutBtn = new JButton("Logout");

donateBtn.setBackground(secondary); donateBtn.setForeground(Color.WHITE);
resetBtn.setBackground(primary); resetBtn.setForeground(Color.WHITE);
logoutBtn.setBackground(new Color(180,0,0)); logoutBtn.setForeground(Color.WHITE);

buttonPanel.add(donateBtn);
buttonPanel.add(resetBtn);
buttonPanel.add(logoutBtn);

donateBtn.addActionListener(e -> {
    if (donorName.getText().trim().isEmpty() || amount.getText().trim().isEmpty()) {
        JOptionPane.showMessageDialog(this, "Please fill required fields", "Error",
JOptionPane.ERROR_MESSAGE);
        return;
    }
    String id = "D" + (DonationManager.donations.size() + 1001);
    Donation d = new Donation(id, donorName.getText(), u_email, phone.getText(),

```

```

        "", city.getText(), (String)stateBox.getSelectedItem(), amount.getText(),
        (String)categoryBox.getSelectedItem(), (String)paymentBox.getSelectedItem(),
        pan.getText(), addressArea.getText(), messageArea.getText());
        DonationManager.donations.add(d);
        JOptionPane.showMessageDialog(this, "Thank You For Your Donation!", "Success",
        JOptionPane.INFORMATION_MESSAGE);
    });

```

```

        resetBtn.addActionListener(e -> {
            phone.setText("");
            city.setText("");
            amount.setText("");
            pan.setText("");
            addressArea.setText("");
            messageArea.setText("");
            genderGroup.clearSelection();
            anonymous.setSelected(false);
            stateBox.setSelectedIndex(0);
            categoryBox.setSelectedIndex(0);
            paymentBox.setSelectedIndex(0);
            donorName.setText(originalName);
            donorName.setEditable(true);
        });

```

```

        logoutBtn.addActionListener(e -> { dispose(); new MyFrame(); });

```

```

        JPanel centerPanel = new JPanel();
        centerPanel.setBackground(Color.WHITE);
        centerPanel.setLayout(new BoxLayout(centerPanel, BoxLayout.Y_AXIS));
        centerPanel.add(formPanel);
        centerPanel.add(addressPanel);
        centerPanel.add(messagePanel);
        centerPanel.add(optionPanel);

```

```

        add(heading, BorderLayout.NORTH);
        add(centerPanel, BorderLayout.CENTER);
        add(buttonPanel, BorderLayout.SOUTH);

```

```

        setVisible(true);
    }
}

```

//LOGIN SCREEN

```

class MyFrame extends JFrame implements ActionListener {
    JPasswordField pass = new JPasswordField(15);
    JPasswordField userEmail = new JPasswordField(15);
    JButton submit = new JButton("Login");

```

```

String u_names[] = {"Super Admin","Gurujyot Singh","Test user"};
String emails[] = {"admin@gmail.com","guru@gmail.com","test@gmail.com"};
String passwords[] = {"admin123","guru123","test123"};
String user_id[] = {"NGOAD01","NGONU01","NGONU02"};

JPanel leftPanel = new JPanel(new BorderLayout());
JPanel rightPanel = new JPanel();

MyFrame() {
    try {
        setIconImage(new
ImageIcon("C:\\xampp\\htdocs\\practicefolder\\ANANDSEWA\\LOGO.png").getImage());

        JLabel label = new JLabel("hi");
        label.setForeground(Color.white);
        BoxLayout box = new BoxLayout(rightPanel,BoxLayout.Y_AXIS);
        rightPanel.setLayout(box);

        ImageIcon icon = new ImageIcon("C:\\xampp\\htdocs\\practicefolder\\ANANDSEWA\\hero.jpg");
        label.setIcon(icon);
        label.setHorizontalTextPosition(JLabel.CENTER);
        label.setVerticalTextPosition(JLabel.TOP);
        leftPanel.add(label,BorderLayout.EAST);

        rightPanel.add(Box.createVerticalGlue());
        JPanel userRow1 = new JPanel(new GridLayout(1,2));
        JPanel userRow2 = new JPanel(new GridLayout(1,2));
        JLabel message = new JLabel("LOGIN FORM");
        message.setFont(new Font("Arial", Font.BOLD, 30));
        message.setHorizontalAlignment(SwingConstants.CENTER);
        message.setAlignmentX(Component.CENTER_ALIGNMENT);

        JLabel email = new JLabel("Enter Your Email : ");
        email.setFont(new Font("Poppins",Font.BOLD,17));
        userRow1.add(email);
        userRow1.add(userEmail);

        JLabel password = new JLabel("Enter your password : ");
        password.setFont(new Font("Poppins",Font.BOLD,17));
        userRow2.add(password);
        userRow2.add(pass);

        userEmail.setBorder(null);
        pass.setBorder(null);
        submit.setBorderPainted(false);
        submit.setContentAreaFilled(true);

```

```

submit.setBackground(new Color(0, 120, 215));
submit.setForeground(Color.WHITE);
submit.setFont(new Font("Arial", Font.BOLD, 16));
submit.setPreferredSize(new Dimension(150, 40));

userRow1.setMaximumSize(new Dimension(Integer.MAX_VALUE, 30));
userRow2.setMaximumSize(new Dimension(Integer.MAX_VALUE, 30));

rightPanel.setBorder(BorderFactory.createEmptyBorder(20, 20, 20, 40));
rightPanel.add(message);
rightPanel.add(Box.createVerticalStrut(60));
userRow1.setOpaque(false);
userRow2.setOpaque(false);
rightPanel.add(userRow1);
rightPanel.add(Box.createVerticalStrut(20));
rightPanel.add(userRow2);
rightPanel.add(Box.createVerticalStrut(20));
rightPanel.add(submit);
rightPanel.add(Box.createVerticalGlue());

this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
this.setTitle("Anand Sewa NGO");
this.setLayout(new GridLayout(1,2));
this.add(leftPanel);
this.add(rightPanel);
leftPanel.setBackground(new Color(30,23,96));
rightPanel.setBackground(new Color(255, 172, 12));
this.setSize(1000,800);
this.setVisible(true);

submit.addActionListener(this);
} catch (Exception ex) {
    ex.printStackTrace();
}
}

public void actionPerformed(ActionEvent e) {
    if (e.getSource() == submit) {
        String u_email = userEmail.getText();
        String password = String.valueOf(pass.getPassword());
        String userID = null;
        String userName = null;
        boolean isUser = false;

        for (int i = 0; i < emails.length; i++) {
            if (u_email.equals(emails[i]) && password.equals(passwords[i])) {
                isUser = true;
            }
        }
    }
}

```

```

        userID = user_id[i];
        userName = u_names[i];
        break;
    }
}

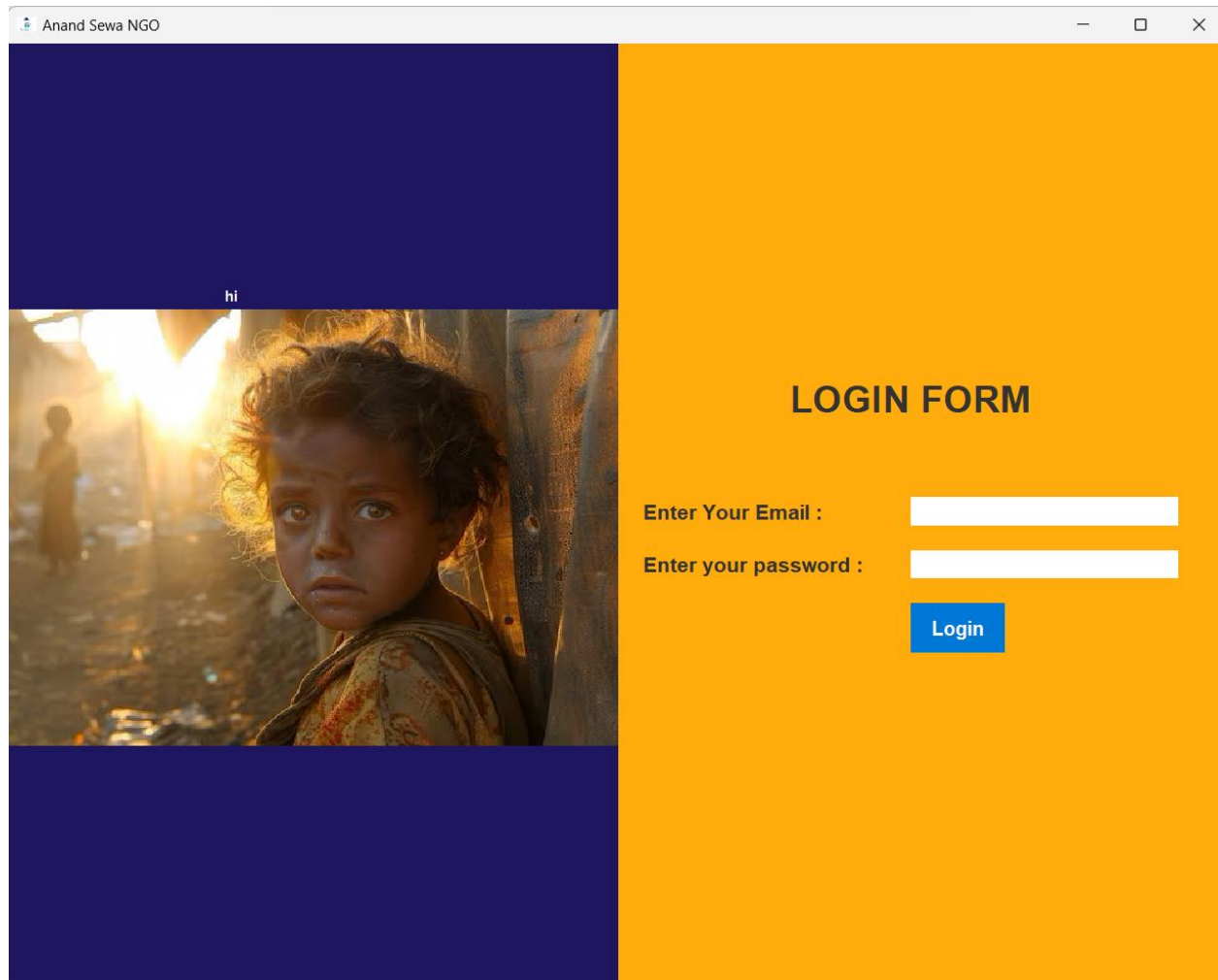
if (isUser) {
    this.dispose();
    if (userName.equals("Super Admin")) {
        new AdminDashboard();
    } else {
        new HomeScreen(userName, u_email, userID);
    }
} else {
    JOptionPane.showMessageDialog(this, "Invalid email or Password!\nPlease try again.", "Login
Failed", JOptionPane.ERROR_MESSAGE);
}
}
}

class JFrameDemo {
    public static void main(String args[]) {
        new MyFrame();
    }
}

```

Results

5.a. Output Screenshots



Anand Sewa NGO - Donation Form

ANAND SEWA NGO DONATION PORTAL

User ID

NGONU01

Donor Name

Gurujyot Singh

Email

guru@gmail.com

Phone Number

Gender

☐ Male
☐ Female
☐ Other

City

State

Punjab

Donation Amount

Donation Category

Education Support

Payment Method

UPI

PAN Number

Address

Purpose / Message

☐ Donate Anonymously
☐ Receive NGO Updates

Donate Now
Reset Form
Logout

Anand Sewa NGO - Donation Form

ANAND SEWA NGO DONATION PORTAL

User ID

NGONU01

Donor Name

Gurujyot Singh

Email

guru@gmail.com

Phone Number

9843784573

Gender

☒ Male
☐ Female
☐ Other

City

Bathinda

State

Punjab

Donation Amount

3000

Donation Category

Education Support

Payment Method

UPI

PAN Number

7845765432

Address

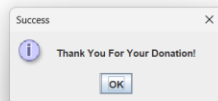
asdfsdf

Purpose / Message

asdfsdfs

☐ Donate Anonymously
☐ Receive NGO Updates

Donate Now
Reset Form
Logout



5.b. Description of Outputs

The login screen successfully ask for email and password and block wrong users who don't know the password.

- The Admin Dashboard show a neat list of all people who give money, with their name and amount clearly visible.
- The Donation Form successfully take all the details and show a 'Thank You' popup when saved, which make the user feel good.
- It work very good and smooth without any freezing or hanging of the computer.

Testing and Validation

6.a. Test Cases

We test the program many times to make sure it not break when user do something wrong or silly.

6.a.i. Define and describe test cases used for validating the code

- **Test Case 1:** Leave Donor Name and Amount blank and click Donate.

Expected Result: Show error popup 'Please fill required fields'.

Actual Result: Error popup showed correctly and stop the empty data.

- **Test Case 2:** Put wrong password in login screen.

Expected Result: Show 'Invalid email or Password!'.

Actual Result: Error message showed correctly and don't let the user inside.

- **Test Case 3:** Click 'Update' without selecting any record in admin.

Expected Result: Show 'Please select a record to update'.

Actual Result: Worked as expected and don't crash the program.

6.b. Validation

We check if the program do exactly what we tell it to do in the code.

6.b.i. How outputs were verified against expected results

We check the output of the program with what we expect it to do before running it. The error messages come exactly like we code it, and the data successfully add to the list and show on the screen. So, the outputs are validate and the program is working correct and safe to use.

Challenges Faced

7.a. Technical Challenges

I face some problems while making this project in the computer lab.

7.a.i. Issues faced during project

- Sometime I forget to put the semicolon ';' at the end of the line.
- The computer show a big red squiggly line and I get scared that I break the code.
- Also, the GridLayout was making the txt boxes too small to read the text inside.
- The image path for the NGO logo was giving file not found error at first because of double slashes.
- My computer was sometime slow and take long time to compile the Java code.

7.b. Resolutions

7.b.i. Steps taken to overcome the challenges

- To solve the semicolon problem, I read the error message at the bottom carefully and add the ';' where it say.
- To fix the layout, I ask my teacher and read on internet, and I change the grid size and add some empty borders to make it look good.
- I also fix the image path to the correct folder so the logo show up nice on the screen.
- I close other heavy programs like Chrome to make my computer run the Java code faster.

Conclusion

8.a. Summary of Findings

In the end, the project is finish and it work very well. It successfully help the NGO to save time and stop them from using ld papr books. The data is safe in the computer memory and can be find very fast.

8.a.i. Briefly summarize the project outcomes

The software is fly working and do all CRUD (Create, Read, Update, Delete) operations good.

The GUI is colorful and easy to understand for any person.

8.b. Learning Outcomes

I learn alot of new and exciting things in this project.

8.b.i. Describe the skills or knowledge gained

- I lear how to make classes and objects in Java and how they talk to each other.
- I learn how t use Swing GUI tomake windws, buttons, and text fields.
- I lear how to use eent handling to make buttons do things when clicked by mouse.
- I learn how to use ArrayList to store data in memory and loop through it.
- I learn how to read error messages and fix my wn mistakes without crying.

8.c. Future Scope

We can make itven better in the future with more features.

8.c.i. Discuss how the work could be extended or applied in real-world scenarios

In the future, we can connect this software to a real MySQL database. Right now, if we turn off the computer, the data in ArrayList will delete because it is only in RAM. If we use MySQL, the data will save forever on the hard disk. We can also make a website version of it or a mobile app so people can donate from their phone anywhere in the world. We can also add a feature to print a receipt for the donor.

References

9.a. Cite any books, articles, or online resources referred to during the project

- Oracle Official Java Documentation website (docs.oracle.com).
- W3Schools Java and Swing Tutorial for beginners.
- Class notes and guidance from our respected teacher Dr. Bhavin Shah.
- Youtube videos about Java JFrame, ActionListener, and GridLayout.